

The etcGEM work: what has been established, what is missing, and what the evidence supports

A shared record for the group, assembled 2026-09-09

Gabriel Yvon-Durocher

Living Systems Institute, University of Exeter, Stocker Road, Exeter EX4 4QD, UK

2026-09-09

Table of contents

1	How to read this document	1
2	Where things stood	2
3	The merge, and what the gates prove	2
4	What the models turned out to be doing	3
5	What the predictors turned out to be doing	3
6	What sets a thermal curve in these models	4
7	The arithmetic behind the <i>Candida</i> figure	4
8	The membrane-area mechanism	5
9	Sampling, and a quantity that is not well defined	6
10	Defects this project introduced	6
11	What is missing	7
11.1	Measurement gaps — no amount of modelling closes these	7
11.2	Model gaps — mechanisms the framework cannot express	7
11.3	Housekeeping — known, bounded, and costed	7
12	The publication assessment	8
12.1 (a)	The <i>Candida</i> thermal-limit paper	8
12.2 (b)	The <i>E. coli</i> gas-flux and overflow work	8
12.3 (c)	The cross-organism activation-energy comparison	9
12.4 (d)	A methods paper on what enzyme-constrained thermal models can and cannot do	9
13	How to update this document	10

1 How to read this document

This is a record of where the enzyme- and temperature-constrained genome-scale model (etcGEM) work stands, written for the three of us. It covers the **framework** — the shared core, the seven organism models on

it, and what has been learned by testing them. Empirical work appears only where the framework consumes it: measured thermal performance curves as calibration or validation targets, the measured meltome as a predictor benchmark, measured activation energies as comparators. The Candida manuscript’s own empirical results are not ours to assess and are not assessed here.

Every number in this document comes from a file. `evidence.csv` in this directory carries one row per claim: the value, the file, the commit that last wrote that file, and whether the claim is CURRENT, HISTORICAL (true of an earlier model state) or PROVISIONAL-ON-P6. Claims are cited by their row id, like [W1]. Nothing is quoted from memory or from another summary.

Several of the sharpest findings came from correcting things this project itself got wrong. Those are in §10, with the same prominence as everything else. That section is not throat-clearing: two of the seven changed a conclusion.

A caution that applies throughout. A large fraction of this week’s results are **negative** — things that do not work, do not generalise, or turn out to be arithmetic. Some negatives are valuable because they close a question or invalidate a comparison; others are simply absent effects. §12 distinguishes them where it matters, and does not inflate either kind.

2 Where things stood

Two implementations of the same idea existed side by side [W1]. `src/etcgem` carried three organisms — *E. coli*, *M. maripaludis*, *Synechocystis* — with a config model, providers, one thermal layer and one enzyme-cost layer [W2]. Ilgaz’s `gem/` carried four *Candida* species with its own thermal function, its own path map and species handled by a dictionary. Parsa’s six gas-exchange configurations sat on a July snapshot of the core with new modules beside it.

Nothing was wrong with any of them individually. The problem was that the method existed in three places, so a change in one could not be checked against the others, and the published *Candida* result rested on numbers whose absolute scale nobody could reproduce from the shared code.

3 The merge, and what the gates prove

Both bodies of work are now on one core. Four *Candida* strains are `strains/` folders carrying data and configuration and no code [W3]; Parsa’s configurations are `configs/experiments/` overlays and one CLI verb. Seven organism models share one thermal layer, one enzyme-cost layer and one config model.

Three gates hold that together, and all three are re-run at every merge:

gate	what it proves	result
K1	the <i>Candida</i> port reproduces the standalone implementation	79/79 PASS [G1]
P1	configurations A, B and C reproduce Parsa’s committed gas-flux numbers	60/60 PASS , worst difference 8.3×10 [G2, G3]
P3	configurations D, E and F reproduce the R^2 his reports print	10 of 10, worst 0.009 [G4]

The K1 gate reproduces the standalone *including a ~0.4 % error its own solver made* at two points, which is why the repository now distinguishes **fidelity to the original** from **numerical correctness** and keeps both configurations. His v3 posterior is identical to the one committed here to every stored digit [G5], so the two codebases share a calibration and not just a method.

What a gate does not prove. It checks the fields it compares. The P3 gate evaluates one parameter point once, after a fresh build, on both sides — as a *port* check it stands. It says nothing about whether the quantity being scored is well defined, which §9 returns to.

4 What the models turned out to be doing

The most consequential class of finding this week was not about temperature at all. It is that these reconstructions contain **uncosted shortcuts**: routes that carry flux without paying the enzyme cost the whole method is built on. Four instances, three organisms, one of them in a published reconstruction.

E. coli: four O sinks. Four O⁻-consuming side reactions carry ~zero enzyme cost and short-circuit the electron transport chain, creating a futile cycle that was ~75 % of gross O⁻ turnover at LB/40 °C. Closing them costs ~0.3 % growth and is now a provider default [S1]. This was Parsa’s audit, and it is the template for the rest.

The three *Candidozyma*: a free proton circuit. Their respiratory chain supplies **0.04–0.05 %** of the protons ATP synthase consumes; *C. parapsilosis* supplies 112 % and *E. coli* 124 % [S2]. Eleven uncosted, **reversible** metabolite/H⁺ symporters close the circuit outside the chain. The separating feature is reversibility, not uncostedness: *E. coli* has twelve uncosted proton movers and none reversible [S3]. Constraining the direction repairs it at a cost of 36–38 % of predicted growth [S4]; costing the transport does not. The defect is in the published *iRV973*, verified read-only, and is shared by the three models derived from it.

And a second defect in the same scaffold: complex III moves its protons cytosol→matrix, the opposite direction to a mitochondrion, so it *spends* the gradient rather than building it [S5].

The audit had a blind spot, and it was structural. `etcgem` audit-sinks watched chemical carriers and terminal acceptors — classes A to D. **None watched the ion ATP synthase runs on** [S6]. A free proton circuit passed an audit that reported 44–47 class-A hits in the same models. Class E now covers it, and getting it right required three things that would each have mis-diagnosed an organism: inferring the coupling ion per strain (*M. maripaludis* is sodium-driven [S7]), counting chemistry as well as translocation (*Synechocystis* scores 19 % on translocation and 100 % with the lumen’s water splitting counted), and handling GECKO’s arm/isozyme split (a naive test finds no ATP synthase in *eciML1515* at all).

What this means for the method, stated carefully. An enzyme-constrained model’s central claim is that flux costs protein. A route that carries flux for free is not a modelling approximation; it is a hole in that claim, and it is invisible to every check that looks at growth. Four have now been found by looking, and the fourth was found only because a class of check did not exist. That is a statement about how these reconstructions are built and audited, not about any one of them.

5 What the predictors turned out to be doing

Every species-specific input the *Candida* etcGEM has comes from sequence predictors. A1 tested the load-bearing one against measurement.

Seq2Tm resolves thermostability between organisms and not within a proteome. Across the tree of life, $r = +0.762$ [A1]. Within one proteome, against 1947 measured *S. cerevisiae* melting points, $r = -0.048$ [A2]. Where it can be checked against a measured congeneric difference it under-states it **17-fold** [A3].

The bias is not uniform, and that matters more than its size. The mean is +5.43 °C [A4], but it rises **1.06 °C per °C of predicted Tm** — from +2.44 °C in the lowest predicted octile to +12.78 °C in the highest [A5]. The cause is A1’s own headline: measured Tm barely responds to the prediction (slope -0.061 [A6]), so almost all of the predicted spread is error, and the error grows with the prediction. A flat offset therefore corrects the *location* of the Tm distribution and not its *shape*.

What could not be obtained. Per-protein measured *S. cerevisiae*/*S. uvarum* Tm; the authors publish the summary only [A8]. So the strongest test — the correlation of measured against predicted *differences* —

cannot be computed, and the congeneric comparison is summary to summary. Nothing was substituted for it.

6 What sets a thermal curve in these models

Three separate constraints have each been shown to relocate the growth optimum, on the same organism and model:

constraint	effect on T_{opt}	effect on CT_{max}
the sector re-grounding (a416fd1)	$37 \rightarrow 30$ °C [T1]	+0.05 %
the translation cap, where it binds	T_{opt} is where the cap starts to bind [T2]	unmoved
a total-carbon cap	$39.0 \rightarrow 32.5$ °C between c_{max} 80 and 60 [T3]	0.66 % [T4]

The pattern is mechanical rather than coincidental. CT_{max} is where the unfolding envelope takes growth to zero, and a capacity constraint can only lower the curve — it cannot move the temperature at which enzymes denature. T_{opt} is wherever the *binding* constraint’s ceiling peaks or crosses the rising envelope-limited branch, so it moves whenever **which** constraint binds changes, with no change to the envelope at all.

The standing rule that follows is now written down: T_{opt} **is quoted with its binding constraint named; CT_{max} may be quoted plainly** [T5]. A variance decomposition that reports “ T_{opt} is owned by protein stability” is a statement about one operating point, measured with local perturbations that do not cross a regime boundary. It is not wrong; it is conditional, and the condition is now stated in the *E. coli* report.

Two structural limitations of the growth envelope, found separately. The model’s growth activation energy is too steep against measurement in every *Candida* species, by **1.24–2.29×**, and the error barely moves between configurations — it is structural, not a tuning failure [T7]. About **88 %** of it is the shared ΔC_p prior [T8]; a value inside the literature range reconciles the activation energy *and* improves the fit. But the same parameter **cannot** close the ceiling over-prediction and pushes it slightly the wrong way [T9]: two failures, not one failure with one cause.

And one artefact worth knowing about because it looks like biology: the *E. coli* glucose proteome was measured at 25, 30 and 37 °C only, so above 37 °C the allocation is held constant and the nominal curve is **bit-identical across eight grid points** at 0.524757 h⁻¹ [T6]. The model is answering the same question eight times; the shoulder is the data table running out.

7 The arithmetic behind the *Candida* figure

Figure 4’s claim is a ratio: the T_m separation the model *requires* between relatives, over the separation the predictors say is *available*. Both halves have moved, for different reasons, and the sequence is the finding.

The requirement fell by more than half when the model changed, and has been stable since. 32.54 °C as published [F1]; 13.77 °C once it runs on this repository’s thermal form (K2), and 13.64 °C on this repository’s own recompute of that state; **13.57 °C** after the K5 proton-circuit repair [F3]. It is also **convention-independent** — the two growth conventions differ by 0.00 °C on the repaired model [F4] — because the counterfactual makes zero calls to the predicted strain’s measured curve. That retires a worry about a number already in circulation.

The available separation rose 6-fold when the predictor was corrected, from 0.411 °C to 2.56 °C [1.33, 5.56], and the fold gap with it: **$\sim 79\times \rightarrow 34\times \rightarrow 5.4\times$ [2.5, 10.3]** [A7]. Still a failure, and a materially different sentence.

Two things about the remaining gap. First, applying A1’s measured bias — untuned, at exactly the value A1 measured — closes the ceiling gaps by **a third to a half** in all four species, and the rising limb does not move [F8]. Full closure would need **1.8–2.8×** **that bias, differing by 5 °C between species** [F9],

Figure 4's arithmetic, stage by stage — every value read from a committed table

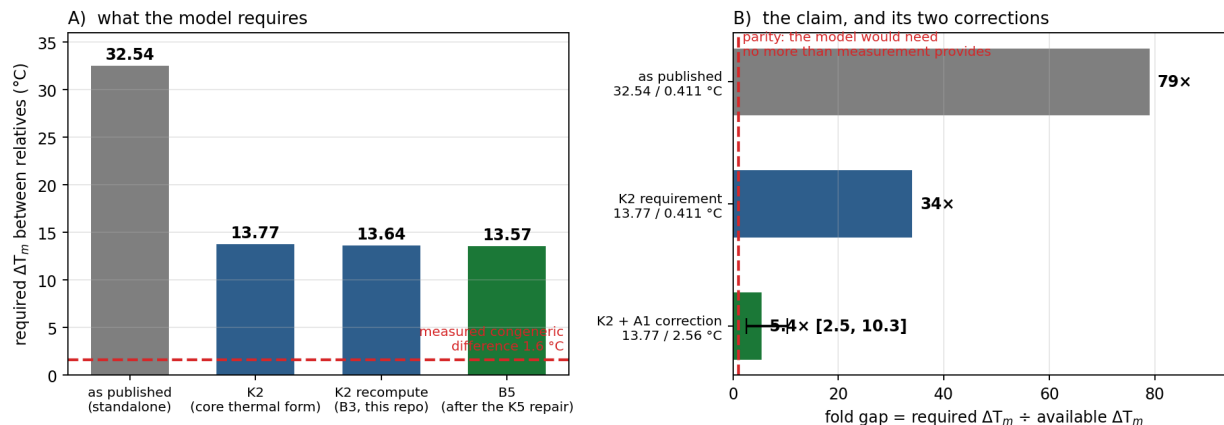


Fig. 1 | Figure 4's arithmetic, stage by stage. **A)** the T_m separation the model requires between relatives, at four model states, against the only measured congeneric benchmark there is (1.6 °C over 827 ortholog pairs, dashed). **B)** the fold gap that constitutes the claim, with A1's two corrections and the credible interval on the corrected value. Every value is read from a committed table; producing script reports/synthesis/fig_requirement_arithmetic.py, sources named in its docstring.

which no uniform predictor bias can produce. Second, and this is the harder constraint: the model's $CT_{\max}$ spans 1.6 °C across the four species while the **observed** limits span 6 °C [F10]. The variation in the gap is almost entirely observational, so no uniform correction can differentially fix the gaps — only their average.

And the interspecies requirement turns out to be arithmetic, not a model constraint [F6]. Recomputing the required per-species offset under five model states, the model's own ceiling spread varies 4.5-fold while the required-offset spread moves by $\pm 3\%$, pinned to the observed 6.00 °C throughout. The interspecies difference is the observed thermal-limit difference wearing a model's clothes, and must not be set beside a measured congeneric difference as though the two were commensurable. What survives is the *per-species* statement — “this model's ceiling is 9.8 °C too high for *C. auris*” is a real model statement — and that is what §12 uses.

The other candidate explanation did not generalise either: the ~5.6 K common ceiling term that looked like it might be shared is a **one-organism observation**, with residuals spanning 0.05–10.48 °C across the strains where they can be computed [F7].

8 The membrane-area mechanism

P1 collapsed Parsa's configurations E and F into one table-driven ETC membrane-area constraint in the core — an area budget $\sum_i (A_i/k_{\text{cat},i}) v_i \leq A_{\text{ETC}}$ driven by a per-strain table, where configuration F is one column of that table rather than a second module. P3 gated it on measured *E. coli* respiration [M1].

K4 carried it to the four *Candida* strains, which is what §6a of the discussion notes had wanted for months. **It cannot carry the interspecies comparison**, for two independent reasons, and neither is the one anticipated [M3]. Structurally, in three of the four models the chain carries no load, so an area budget on ETC flux has nothing to bind on — the same proton-circuit defect as §4, seen from a different direction. Parametrically, no measured fungal ETC footprint or mitochondrial inner-membrane area exists. Re-tested with both escape routes closed, the required area factor is **122-fold** and still points the wrong way [M4].

Also worth recording: **alternative oxidase is in all four proteomes and in none of the four models** [M2] — the one genuinely fungal branch of the ETC, and the mechanism Xiao et al. propose, has no reaction to attach a footprint to.

9 Sampling, and a quantity that is not well defined

No chain in this family is converged, and that applies evenly. The nine refits run 146–245 against 1500–2000 steps — chain/ 6–12 against the 40 criterion [P1]. Measured with the same estimator, **all six of Parsa’s committed chains** give chain/ 8.2–9.4, n_eff 148–170 [P2]. Both families have run about nine autocorrelation times. Reaching the criterion is ~8000 steps, 40 h for all nine [P3]. This is a sampling-budget property, not a failure of model or diligence, and the arithmetic is given so the decision to spend it is a considered one.

And a sharper problem underneath it, found by P6’s pre-flight [EF1] (*on branch p6/convergence; PROVISIONAL-ON-P6 as to location, not as to measurement*). For configurations E and F the log-likelihood **varies by 0.13–7.49 at a fixed parameter vector**, against a posterior width of about 7. The growth term is stable; the **O uptake is not** — it differs by up to 9.4 mmol gDW⁻¹ h⁻¹ on a mean of 19.8 between repeats at the same vector. The cause is identified: the model’s O uptake at optimal growth is **not unique**, and which vertex the solver returns depends on the basis left by the previous solve. Configurations E and F restructure the problem on every call (remove and re-add the area constraint); configuration D does not, and its jitter is 0.0004.

So the “respiration R²” those chains report was scored on a quantity the model does not determine. **This does not touch the P3 gate as a port check** — it evaluates one point once after a fresh build on both sides — and it does not touch configuration D. It does mean that any respiration conclusion from configuration E or F needs the degeneracy resolved first, by fixing the flux distribution (a secondary objective, or parsimonious FBA) rather than by longer chains.

10 Defects this project introduced

Seven, listed because the document’s credibility depends on their being here rather than in a footnote. Two changed a conclusion.

	what	how it surfaced
O1	a default flip broke etcgem tpc outright for every gecko-provider strain with sectors	the next prompt’s own merge verification, before it was pushed
O2	cli.main returned the output path into sys.exit() , so every successful run exited 1	it made a verification pass silently vacuous — cmd && check skipped the run and compared a file with itself
O3	the wrong ETC table was wired as configuration F	the P3 gate; his fits use configuration E’s table plus one column
O4	a gate run left an overlay’s carbon cap on, so E and F silently carried a cap they were never fitted with	it made NLDM look irreproducible (respiration R ² –2.5); caught before any conclusion was drawn
O5	the warm start failed silently in all nine refits	chains started at the emergent point, lengthening burn-in; repaired
O6	K5’s activation-energy sign result did not stand	K6 checked the comparator: two measured activation energies are not the same quantity
O7	a prompt’s own headline table did not survive its test	K9 corrected it: the interspecies requirement is arithmetic

O3 and O6 changed conclusions. O2 is the instructive one: it did not corrupt a number, it disabled a *check*, and the check then reported success. The standing responses adopted are (i) read settings back from the resolved

config rather than assuming them, (ii) never verify with `cmd && check`, and (iii) a generated provenance stamp on every report [R2], because the audit that started this found **8 of 11** input directories no longer reproducing [R1].

11 What is missing

Three kinds, because they need different responses.

11.1 Measurement gaps — no amount of modelling closes these

what is missing	what it would license	cost
a measured meltome for any <i>Candida</i>	replaces the load-bearing predictor with data; would settle §7 directly	a thermal proteome profiling experiment
per-protein <i>S. cerevisiae</i>/<i>S. uvarum</i> Tm	A1's strongest test — the correlation of measured against predicted <i>differences</i>	an email to the authors
mitochondrial inner-membrane area per gDW, per species, at 30 and 40 °C	the measurement an ETC-area budget actually consumes; without it the mechanism is parameterised from <i>E. coli</i>	electron microscopy / organelle fractionation
measured fungal ETC footprints and turnovers	the same, for the table's rows	literature or structure-based estimation
organelle-resolved <i>Candida</i> lipidomics	whole-cell data are rich; fractionated data do not exist	fractionation + lipidomics
the <i>E. coli</i> proteome above 37 °C	removes the bit-identical 37–44 °C shoulder [T6]	extend an existing series
per-cell carbon conversion for the respirometry	the inoculum back-projection is off in every row, and <code>cell_volume_um3/cell_carbon_fg</code> are typed constants that disagree with the pipeline's own log by $\sim 6\times$	a cell-count and dry-weight calibration

11.2 Model gaps — mechanisms the framework cannot express

- **The membrane as a physical structure.** The area budget is a bound on ETC flux, not a representation of a membrane. It cannot express lipid composition, curvature, or the trade-off between respiratory and transport machinery for the same surface.
- **Whatever kills cells below the unfolding ceiling.** The model's $CT_{\max}$ is set by enzyme denaturation, and it over-predicts by 9–14 °C. Applying the measured predictor bias closes a third to a half [F8]; the residual differs by species [F9] and no uniform correction can produce that. Something the model does not contain sets the real limit.
- **Alternative oxidase** [M2], and more generally fungal-specific ETC branches: absent from the reconstructions, so absent from every conclusion.
- **The interspecies term is arithmetic** [F6]. The framework as it stands cannot make a *difference between species* a model prediction rather than a restatement of the observed difference. That is the deepest gap in §7, and it is structural.

11.3 Housekeeping — known, bounded, and costed

- **Under-converged chains** [P1, P2, P3] — ~ 40 h for all nine; and for configurations E and F, longer chains are the *wrong* fix until the flux degeneracy is resolved [EF1].
- **The unexplained 10 °C** — *Synechocystis*' committed ceiling is not reproducible from a strain build (OPEN_ITEMS 3.19); recorded with strain and evidence, not investigated.

- **Stale artefacts** — 8 of 11 directories behind the *E. coli* report [R1]; the standing fix is the provenance stamp, not recomputation, and regeneration is deferred to paper time.
- **resolved_config.yaml records an absolute model_path**, so “byte-identical on re-run” is only checkable from the same directory (OPEN_ITEMS 3.15).

12 The publication assessment

FOR PI JUDGEMENT. This section lays out the evidence for and against each candidate. It reaches no verdict and makes no recommendation on which to pursue. That is deliberate.

12.1 (a) The *Candida* thermal-limit paper

The claim it would make. That the growth-limit difference between *C. auris* and its relatives cannot be explained by the thermostability difference their proteomes carry — the model requires far more separation than is available.

Evidence in hand. The requirement is now stable across model states and repairs: 13.57 °C [F3], convention-independent [F4], and reproduced by an independent implementation [G1]. The predictor validation is quantitative and unusually direct [A1–A3, A5, A6]. The corrected fold gap is $5.4\times$ [2.5, 10.3] [A7] — still a failure of the explanation, on corrected arithmetic.

What is missing. A measured *Candida* meltome. Everything on the “available” side is predicted by a tool shown not to resolve within-proteome variation [A2].

The strongest referee objection. *The interspecies requirement is a restatement of the observed difference, not a model prediction* [F6] — the model’s own ceiling spread varies $4.5\times$ across states while the requirement moves $\pm 3\%$. A referee who sees this will ask what the model contributed. **The defensible form of the claim is per-species:** “this model, with these predicted stabilities, puts *C. auris*’s thermal ceiling 9.8 °C too high, and the predictor bias A1 measured accounts for a third to a half of that” [F8]. That is a real model statement and it survives. The four-species *comparison* is the part that does not.

A second objection to expect. The model’s growth envelope is too steep by $1.24\text{--}2.29\times$ [T7] and the ΔC_p prior accounts for $\sim 88\%$ of it [T8] — a referee will ask why a parameter known to be wrong was not corrected before the requirement was computed. The answer available is that the requirement is insensitive to it (the required-offset spread moves $\pm 3\%$ across $\Delta C_p -3.0$ to -6.0 [F6]), which is a good answer but requires the interspecies framing to be dropped first.

12.2 (b) The *E. coli* gas-flux and overflow work

The claim it would make. That gas exchange and overflow can be made determinate and mechanistic in an enzyme-constrained model, and that a membrane-area budget on the electron transport chain reproduces measured per-cell respiration.

Evidence in hand. Six configurations ported and reproduced [G2, G4]; two independent mechanisms (carbon cap, membrane area) implemented generically and gated; the O-sink correction [S1]; a real measured dataset in two media, ingested with provenance; and configuration D’s growth fits on M9 among the best in the exercise [C8].

What is missing, and it is load-bearing. No chain is converged [P1, P2] — and for configurations E and F the respiration likelihood is not a function of the parameters [EF1], so the respiration R^2 those configurations report is scored on a degenerate quantity. **The membrane-area result is the headline of this work and it is the one directly affected.**

The strongest referee objection. *Your respiration fit is not identifiable.* The answer is not a longer chain; it is to make the flux distribution unique (a secondary objective or parsimonious FBA) and refit. That is a bounded piece of work and it is the single highest-value next step for this line.

A second objection. The medium ceiling K is not identified — posterior/prior width 0.57–0.64 [C6] — so the defined medium’s carbon ceiling is a prior choice, not a fitted quantity. And $c_{\max}$ is medium-dependent in a way that was learned the hard way [C3, C4, C5].

12.3 (c) The cross-organism activation-energy comparison

Read before assessing: `reports/activation_energy/report.qmd`, which predates this week’s work.

The claim it makes. That the ordering methanogenesis > respiration > photosynthesis in thermal sensitivity arises from three distinct cellular routes — where growth control sits on the enzyme E_a distribution, and how much the allocation law buffers it — rather than one mechanism scaled.

Evidence in hand. Three organisms, each calibrated to its own measured curve, each reproducing its observed E_a and the full ordering; a control-weighted decomposition; each organism’s allocation term grounded in its own measured strategy.

Does this week’s work bear on it? Yes, in two specific ways, and one is favourable.

Favourable: the paper quantifies E_a with the **window-independent Sharpe–Schoolfield** E . K6 established independently that this is the valid comparator and that the alternative (an OLS Arrhenius slope through a curve that turns over) is not — the two differ fourfold on growth and not at all on respiration [D1]. The paper already uses the convention that survived a test it did not know it was taking.

Unfavourable, and it needs an answer: E_a is a **rising-limb** quantity, and the rising limb moves with the binding constraint. A carbon cap moved NLDM’s E_a from 0.949 to 1.230 eV — +30 % — between $c_{\max}$ 80 and 60 [T3]. The paper’s three E_a values are each computed at one operating point per organism. **A referee could ask whether the ordering is a property of the organisms or of three particular constraint regimes.** The defence is available and should be made explicitly: each organism is calibrated to its own measured curve, so its operating point is pinned by data rather than chosen; and the decomposition attributes the ordering to *where control sits*, which is a structural property. But the question is now on the record and the paper does not currently address it.

A third, smaller point: the *E. coli* strain in that paper is the same one whose report renders from 8 of 11 directories that no longer reproduce [R1]. Nothing found contradicts a scientific claim — T_{opt} , CT_{max} and curve shape are stable across every test — but if a specific number is quoted it should be regenerated first.

12.4 (d) A methods paper on what enzyme-constrained thermal models can and cannot do

The claim it would make. That this model class has characteristic, findable failure modes, and that the checks currently applied to it do not find them.

The material. Four uncosted-shortcut instances across three organisms, one in a published reconstruction [S1–S5]; an audit blind spot that let a free proton circuit pass a clean audit [S6]; a predictor validated and found to work between organisms and not within a proteome [A1–A3]; the $T_{\text{opt}}/CT_{\text{max}}$ asymmetry with its mechanical explanation [T1–T5]; a likelihood that is not a function of its parameters [EF1]; and a whole family of published and unpublished chains at ~9 autocorrelation times [P1, P2].

Being sceptical of this case, as instructed. There are two ways to read that list.

As one argument: every item is an instance of the same thing — **a model-level claim that no routine check tests.** “Flux costs protein” is falsified by an uncosted route and growth looks fine. “The optimum is set by stability” is falsified by a change of binding constraint and the descriptor still has a value. “The fit is good” is falsified by a degenerate objective and R^2 still computes. Each failure is silent, each is found only by a check designed for it, and the paper would be the set of checks. That is a coherent argument with a clear deliverable, and the class-E audit is proof the checks can be written.

As a list: the items are also just things that went wrong in one week on two codebases. The predictor validation is about a tool, not a model class. The convergence finding is about a sampling budget and is not

specific to thermal models. The T_{opt} result is arguably a restatement of how LP optima work. A referee could reasonably say that three of the five are generic computational-practice observations that happen to have been made here.

The strongest referee objection, if it were written as one argument: *what is the general claim, and is it tested?* The honest answer is that the general claim would be “these failure modes are common and current practice does not detect them”, and this work supports the second half far better than the first — it has one deep instance of each, not a survey. **A survey is what would make it a paper**: run the class-E audit and the shortcut audit across a set of published GEMs and report the incidence. That is a bounded, well-defined piece of work, and it would convert a set of anecdotes into a result.

What is genuinely novel here, and what is not. Novel: the coupling-ion audit class and its finding in a published reconstruction; the within-proteome predictor failure measured against a meltome; the demonstration that a widely-used likelihood is degenerate. Not novel: that GEMs contain artefacts, that MCMC needs to converge, that LP optima are non-unique. The paper’s value would rest on the first three and on the survey it does not yet have.

13 How to update this document

Re-rendering after P6 lands is the intended workflow.

Three claims are **PROVISIONAL-ON-P6** and have defined slots:

row	claim	where it lives	what replaces it
[EF1]	the E/F respiration likelihood is not a function of its parameters	§9, ¶2	P6’s finished report; if the degeneracy is resolved, the sentence changes from “is not” to “was not, and the fix is X”
[C6]	the medium clearance K is not identified	§12	P6’s converged configuration-D posteriors; the width ratio is re-read from <code>compare_posteriors.csv</code>
[C8]	M9 growth R^2 0.959 / 0.987 / 0.982	§12	P6’s converged configuration-D M9 fit; E and F remain blocked by [EF1]

To update:

1. `python reports/synthesis/build_evidence.py` — re-reads every source file’s commit; a row whose source has moved gets the new commit automatically.
2. `python reports/synthesis/fig_requirement_arithmetic.py` — only if a *Candida* requirement changes; it reads `reports/K7_envelope/task1_conventions.csv`.
3. `quarto render reports/synthesis/synthesis.qmd`.
4. `python scripts/stamp_reports.py`.

What must not be done silently. If a number here disagrees with a number in a report, the disagreement is a finding — `evidence.csv` has a `kind=disagreement` row format for exactly that, and there are five already. Record it; do not reconcile it by choosing one.